

University of South Carolina

The Effect of Water Ingestion on Blood Pressure

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BIOL 460L
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November 19, 2024

Objective

Determining the correlation (if any) between ingesting water of varying temperatures and blood pressure immediately following the action.

Hypothesis

There exists a negative correlation between the temperature of ingested water and the change in blood pressure (systolic and diastolic) immediately after ingestion; however, this change levels off to some degree after five minutes.

Finding

There exist positive correlations between the temperature of ingested water and the immediate change in blood pressure (both systolic and diastolic) as well as the delayed change in diastolic blood pressure. The delayed effect on systolic blood pressure correlates negatively with the temperature of ingested water.

Background

Blood pressure is a physiological measure that underlies all cardiovascular processes. Imbalances in blood pressure, therefore, primarily affect the cardiac, pulmonary, and circulatory systems and lead to poor health outcomes. When blood pressure is chronically high, the risk for heart disease increases as hypertrophy occurs in the myocardia. On the other hand, chronically low blood pressure is associated with ischemia since bodily tissues do not receive adequate amounts of nutrients, namely oxygen, causing functional failure (Daixang et al., 2019). Imbalances can occur in systolic and/or diastolic blood pressure, producing several different forms of hypertension and hypotension. When these states become chronic, medical intervention is necessary, often requiring medication-based therapy.

Alternatively, tea is known to have relaxing and blood-pressure-reducing qualities, and the simple drink has been indicated throughout global history as a therapeutic resource for treating blood pressure imbalances and heart disease. Though such specific medical determinations were mostly absent before the twentieth century, the common symptoms of blood pressure imbalances are well-known: lightheadedness, headaches, fatigue, and many others. While numerous types of tea exist with vast effects, the common benefit of blood pressure stabilization draws into question whether its relaxing effects are due entirely to the natural products (and the compounds contained in such products) or to the water used to make the tea. Likely, the natural products used in tea are the primary sources of relaxing qualities, including the reduction of blood pressure (Li et al., 2019). However, water temperature may play a role in both activating compounds in the natural products used to make tea, as well as in affecting the cardiovascular system through thermogenic mechanisms (Jordan, 2005). The overall role of tea in managing cardiovascular health is complex and extends well beyond blood pressure regulation.

In using tea to benefit cardiovascular health, with a specific focus on its effect on blood pressure, the delivery methodology is important. People drink teas at different temperatures, but the most effective brewing and serving temperatures may differ. According to Safdar et al., a hot brewing temperature (80 °C) allows for the most efficient extraction of beneficial phytochemicals from the tea leaves (2016). Drinking tea at this temperature is inadvisable for obvious reasons; blood pressure may skyrocket in reaction to the excessive heat introduced to the body. Then, what drinking temperature conveys the greatest cardiovascular benefit—for example, lowering high blood pressure? The present experiment details the methodology for testing that question by sampling for the effects of ingesting water at different temperatures on systolic and diastolic blood pressure.

Materials and Methods

Researchers began this simple experiment by measuring baseline blood pressures for three participants, using an automatic blood pressure cuff. These and all other results are recorded in a spreadsheet. Meanwhile, a researcher prepared 200 mL water samples of three temperatures: cold, room temperature, and hot. The researcher used ice to prepare the cold water and a hot plate to prepare the hot water. The cold water was standardized with a thermometer to 7.5 °C, the room temperature to 20.5 °C, and the hot to 45 °C.

After the initial blood pressures were taken and water samples created, participants ingested the cold-water sample. Researchers immediately measured blood pressure for each participant and recorded the values. Then, the participants remained seated and breathed normally for five minutes. At this time, the researchers gathered blood pressure measurements again, noting the values for the delayed (second) reading in the spreadsheet. Researchers repeated these same steps for the room-temperature and hot water trials. At the completion of all three trials, the researchers assorted the data, separating by type (systolic or diastolic) as well as by time of reading (immediate or delayed). Four groups are, thus, created to prepare for data analysis: immediate systolic, immediate diastolic, delayed systolic, and delayed diastolic measurements.

To test the hypotheses for this experiment, the researchers calculated the correlation between blood pressure (at both measuring times) and the temperature of the water ingested by participants. Furthermore, the researchers completed two-factor ANOVA tests for each group of data to determine the statistical significance of any correlations determined.

Results

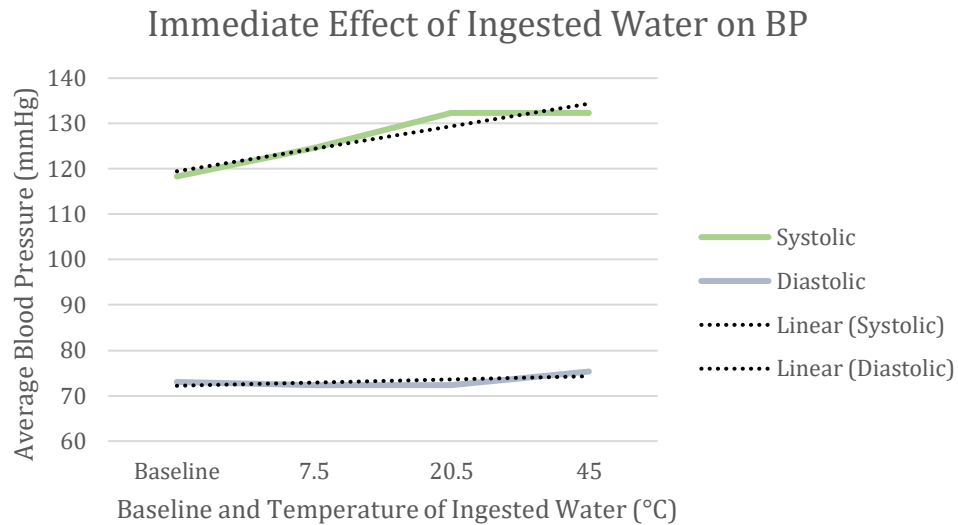


Figure 1: Average blood pressure measured immediately after ingesting water of a certain temperature. Both systolic and diastolic blood pressure generally increase as the temperature of ingested water increases. Dotted lines represent the lines of best fit, indicating positive correlations between variables.

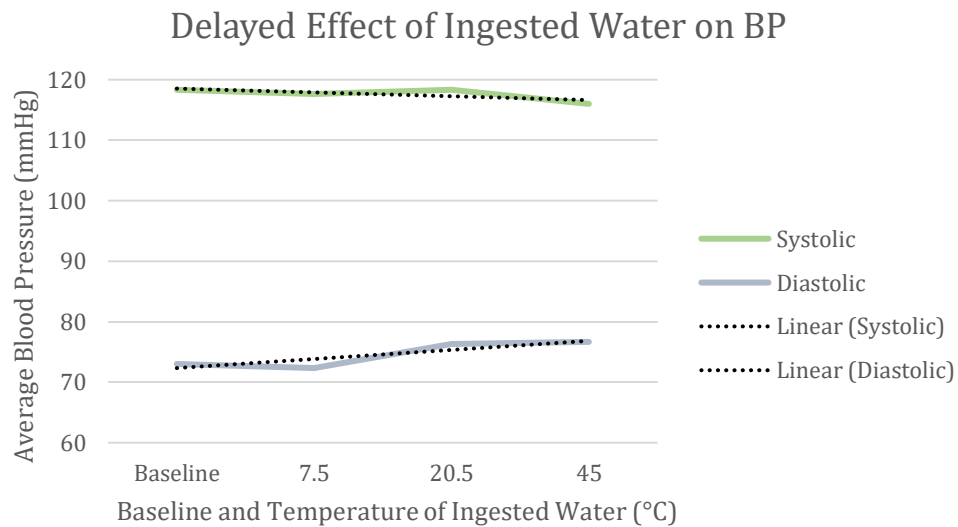


Figure 2: Average blood pressure measured five minutes after ingesting water of a certain temperature. Systolic blood pressure generally decreases while the temperature of ingested water increases. Dotted lines represent the lines of best fit, indicating a negative correlation between systolic blood pressure and the temperature of ingested water but a positive correlation for diastolic blood pressure.

In the first two figures, one might notice the distinct correlations between blood pressure and the temperature of ingested water. First, systolic and diastolic blood pressure immediately increase with temperature, underlying positive correlations between the two variables. However, delayed systolic blood pressure decreases as the temperature of ingested water increases, and these variables are negatively correlated. The change in diastolic pressure for each trial is low relative to the baseline reading. Considering the delayed effect on diastolic blood pressure, pressure increases as the temperature of ingested water increases. Altogether, the data displays that blood pressure does change in response to the ingestion of water (of variable temperature).

Table 1: Correlations, including their strength and direction, between the temperature of ingested water and the consequent change in blood pressure. Values of $r \geq |0.8|$ are considered very strong correlations. Values of $|0.6| < r < |0.8|$ are considered strong correlations.

Trial	Correlation Coefficient (r)	Correlation Strength and Direction
Immediate Effect on Systolic BP	+0.766	Strong, Positive
Immediate Effect on Diastolic BP	+0.940	Very Strong, Positive
Delayed Effect on Systolic BP	-0.808	Very Strong, Negative
Delayed Effect on Diastolic BP	+0.808	Very Strong, Positive

Regarding these correlations between temperature and blood pressure, as displayed in Table 1, correlation strength is generally very strong and positive. Although, the delayed effect on systolic blood pressure is negative yet still very strong in terms of its correlation with temperature. The immediate effect on systolic blood pressure is slightly weaker in correlation yet still categorized as strong. However, these relationships are not statistically significant ($p=0.37$) due to the small sample size in this study.

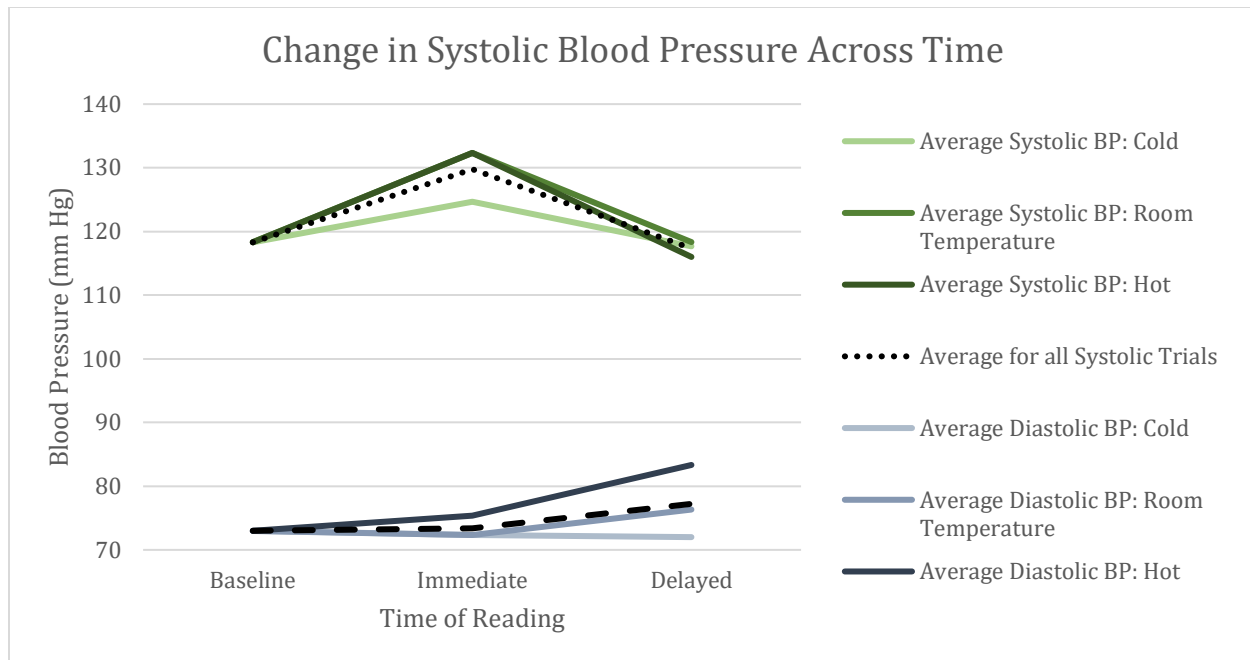


Figure 3: Plot of the changing blood pressures across time. Dotted and dashed (systolic and diastolic, respectively) lines are curves of averaged blood pressure values across all three reading times. The systolic curve of best fit indicates an initial positive correlation between blood systolic pressure and the act of drinking water, but this change reverts within five minutes. The diastolic curve of best fit indicates a longer-term positive correlation between diastolic pressure and the act of drinking water.

Figure 3 exhibits the longitudinal effect of water ingestion on blood pressure. Compared to baseline blood pressure, both systolic and diastolic blood pressure generally increase between the time of drinking water and measuring blood pressure. Yet, the major difference between the effects on systolic and diastolic blood pressure occurs between the first and second measurements. The second measurement (delayed) is taken five minutes after the first. Systolic blood pressure quickly stabilizes and returns to near-baseline by the five-minute mark. On the other hand, diastolic blood pressure more sharply increases (generally) between the first and second measurements. The above results are not statistically significant due to a small sample size ($n < 15$), and an ANOVA test of the relevant data provides confirmation: $p = 0.36$.

Discussion

Contrary to the hypothesis, this study determined that drinking water at any temperature will cause an immediate rise in systolic blood pressure. In terms of diastolic blood pressure, a change (rise) occurs initially but to a minuscule extent. This finding likely relates to the idea that muscular activity increases blood pressure since active muscles require more blood flow. In this case, the swallowing muscles of the mouth, pharynx, and esophagus are activated when drinking water (Sasegbon and Hamdy, 2017). More specifically, drinking cold water caused a lesser increase in systolic and diastolic blood pressure than did drinking room temperature or hot water. Experimentation uncovered a

positive correlation between blood pressure and the ingestion of water at certain temperatures. Generally, drinking water of higher temperatures caused a greater immediate effect on systolic and diastolic blood pressure than did drinking water of lower temperatures.

The delayed effect on systolic and diastolic pressure presented differently, however. While a similar relationship existed between ingestion temperature and the delayed change in diastolic blood pressure, delayed systolic blood pressure decreased toward the baseline. Systolic blood pressure *decreased* at a rate strongly correlated to the temperature of the ingested water. Yet, diastolic blood pressure continued to *increase* at a rate strongly correlated with temperature of the ingested water. Ingesting cold water even *decreased* diastolic blood pressure by a slight amount after five minutes.

These findings indicate an intricate relationship between drinking water and blood pressure. While the researchers had expected systolic and diastolic blood pressures to change concomitantly, distinct relationships arose for the respective pressure types. Such relationships are useful in everyday life, especially for those who struggle with blood pressure disorders. The findings of this experiment suggested that intaking cold water is less detrimental in the short term to blood pressure than intaking warmer water. When blood pressure is high, many people have turned to a warm cup of tea to soothe and relax their bodies. Irrespective of the benefits of the components of teas, colder teas may convey more blood pressure reduction, in that, cold water was less likely to significantly spike systolic or diastolic blood pressure.

Unfortunately, drinking tea at colder temperatures may decrease the bioavailability and effectiveness of the leaves themselves, which limits the health benefits of the compounds (primarily phytochemicals) contained in tea leaves (Safdar et al., 2016). A workaround may include preparing tea at hot, near-boiling temperatures in order to maximize the bioavailability of the phytochemicals, and then serving the tea over ice. Thereby, one can maximize the yield of these beneficial compounds while minimizing the immediate spike in systolic and delayed spike in diastolic blood pressure.

The researchers in the present experiment minimized error by ensuring that successive trials allowed sufficient time for the blood pressure of participants to reset to baseline values. Some errors may exist in the measurement of the temperature of ingested water since slight disparities may have existed between the temperatures of the water each participant ingested. Due to the large range of tested water temperatures, however, this error should be negligible. Another source of error includes the non-standardization of the “immediate” blood pressure measurement. Researchers may have collected these first readings at slightly different times for each participant; though, these differences vary by ten to fifteen seconds at most. Regardless, the possibility exists that these disparities have contributed to some degree of error in data collection. In terms of systematic error, medical test anxiety may have skewed blood pressure measurements since the process of measuring blood pressure represents a clinical test (to some degree). This error is likely to have affected all data similarly—thereby, the data remains reliable. In future experimentation, researchers should standardize the time of the “immediate” blood pressure measurement as well as make larger batches of each water sample so that the temperature holds at the desired value.

Conclusion

The present experiment served to exhibit that both water temperature and the act of drinking water (itself) have profound impacts on blood pressure. Drinking cold water raises systolic blood pressure immediately after drinking, but systolic pressure quickly returns to baseline. Diastolic pressure generally decreases for several minutes after drinking cold water. Drinking room temperature or hot water also raises systolic blood pressure immediately after drinking, and this value returns to baseline within five minutes. These higher temperatures of water gradually increase diastolic blood pressure, however—a stark contrast from drinking cold water. In general, the severity of these effects on blood pressure increases with temperature, but the time-wise relationships (represented in Figure 3) are similar. The data from this experiment are helpful for understanding the best temperature at which to drink tea when using the natural remedy to lower blood pressure. Given that hot-brewed tea maximizes phytochemical output (Maufrais et al., 2018), to further maximize the blood-pressure-reducing capabilities of tea, one should serve it over ice.

In the context of similar experiments, the present research provides a limited scope of the short-term physiological effects that water ingestion employs at various temperatures. In a similar experiment, researchers studied how blood pressure changes depending on the volume of water ingested. This group found that intaking a large volume of water in a short time span significantly increased both systolic and diastolic blood pressure by increasing blood volume (Madhavulu et al., 2014). In a future experiment, the component of volume would allow researchers to determine, even further, how to make the perfect, most heart-healthy cup of tea. Finding the optimal volume at which a sufficient amount of phytochemicals can be delivered while minimizing the volume of water used may serve to accomplish that goal.

To further improve the present experiment, other than by recruiting many more participants, researchers should gather more blood pressure measurements within the five-minute period and collect measurements well past that mark. By extending the experiment past five minutes per trial, one can attain a better longitudinal illustration of the effect of water drinking on blood pressure. Particularly, the present experiment yielded that diastolic blood pressure decreased at the five-minute mark for cold water ingestion and increased for warm and hot water ingestion. By extending the measurement period, one could determine the fullest extent to which the ingestion changed diastolic blood pressure as well as when that pressure returns to baseline. The present study uncovered relevant information regarding blood pressure management, and its continuation could yield more precise and informative data.

References

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